1. Write a program to print all elements of a list using a for loop.
2. Find the **sum of all elements** in a list using a loop.
3. Count how many elements are **greater than 10** in a list.
4. Print all **even numbers** from a list.
5. Print all **odd numbers** from a list.
6. Find the **largest element** in a list without using max().
7. Find the **smallest element** in a list without using min().
8. Count the **number of positive and negative** numbers in a list.
9. Print elements at **even index positions**.
10. Print elements at **odd index positions**.
11. Check whether a given element exists in a list.
12. Count the **frequency of a specific element** in a list.
13. Separate **even and odd numbers** into two different lists.
14. Create a new list containing **only unique elements**.
15. Remove all **negative numbers** from a list.
16. Find the **second largest element** in a list.
17. Reverse a list using a loop.
18. Check if a list is **sorted in ascending order**.
19. Count how many numbers are **divisible by both 3 and 5**.
20. Replace all **negative numbers with 0**.
21. Find the **sum of elements at even indices**.
22. Find the **product of all non-zero elements**.
23. Count how many elements are **prime numbers**.
24. Rotate the list **left by 1 position**.
25. Rotate the list **right by 1 position**.
26. Check if a list is a **palindrome**.
27. Remove duplicate elements **without using set()**.
28. Find the **longest increasing subsequence (basic logic)**.

29. Count pairs whose **sum is even**.
30. Replace every element with the **difference between max and that element**.
31. Print elements that appear **more than once**.
32. Find the **first repeating element**.
33. Count the **number of zeros** and move them to the end.
34. Separate **positive, negative, and zero** elements.
35. Check if all elements in a list are **unique**.
36. Print the list after removing elements at **prime indices**.
37. Find the **majority element** (appears more than n/2 times).
38. Count elements whose **digits sum is even**.
39. Print only those elements whose **square is even**.
40. Replace each element by the **next greater element** (basic approach).

Below are **MCQs** and **Pattern programs using lists** (exam-oriented, Python).
No advanced libraries—only **lists, loops, and conditionals**.

📘 MCQs on Lists (with Answers)

1. What is the output?

lst = [1, 2, 3]

lst.append([4, 5])

print(len(lst))

A) 3
B) 4
C) 5
D) Error

✅ **Answer:** B

---

2. What will be printed?

lst = [10, 20, 30]

print(lst[-1])

A) 10
B) 20
C) 30
D) Error

✅ **Answer:** C

---

3. Which statement removes the first occurrence of 5?

A) del lst[5]
B) lst.pop(5)
C) lst.remove(5)
D) lst.clear(5)

✅ **Answer:** C

---

4. Output?

lst = [1, 2, 3]

lst = lst * 2

print(lst)

A) [1,2,3,1,2,3]
B) [2,4,6]
C) Error
D) [1,2,3]

✅ **Answer:** A

---

5. What does list(range(1,6)) produce?

A) [1,2,3,4,5,6]
B) [1,2,3,4,5]
C) [0,1,2,3,4,5]
D) Error

✅ **Answer:** B

---

6. Output?

lst = [5, 3, 1]

lst.sort()

print(lst)

A) [5,3,1]
B) [1,3,5]
C) Error
D) [3,1,5]

✅ **Answer:** B

---

7. Which creates a copy of a list?

A) new = lst
B) new = lst.copy()
C) new = lst.append()
D) new = list.append(lst)

✅ **Answer:** B

---

8. Output?

lst = [1, 2, 3, 4]

print(lst[1:3])

A) [1,2,3]
B) [2,3]
C) [2,3,4]
D) [1,2]

✅ **Answer:** B

---

9. How many times will the loop run?

lst = [10, 20, 30]

for i in lst:

    pass

A) 1
B) 2

C) 3
D) Infinite

✅ **Answer:** C

---

10. Output?

lst = [1, 2, 3]

print(lst.pop())

A) 1
B) 2
C) 3
D) Error

✅ **Answer:** C

🔷 **Patterns Using Lists (with Code)**

Pattern 1: Print List Elements in Triangle

List: [1, 2, 3, 4]

1

1 2

1 2 3

1 2 3 4

```
lst = [1, 2, 3, 4]

for i in range(1, len(lst)+1):

    for j in range(i):

        print(lst[j], end=" ")

    print()
```

Pattern 2: Reverse Triangle

1 2 3 4

1 2 3

1 2

1

```
lst = [1, 2, 3, 4]
for i in range(len(lst), 0, -1):
    for j in range(i):
        print(lst[j], end=" ")
    print()
```

Pattern 3: Repeated Element Pattern

List: [5, 6, 7]

5

6 6

7 7 7

```
lst = [5, 6, 7]
for i in range(len(lst)):
    for j in range(i+1):
        print(lst[i], end=" ")
    print()
```

Pattern 4: Index-Based Pattern

0

1 1

2 2 2

```
lst = [10, 20, 30]
for i in range(len(lst)):
    for j in range(i+1):
        print(i, end=" ")
    print()
```

Pattern 5: Even Elements Only

List: [1, 2, 3, 4, 5, 6]

2

2 4

2 4 6

```
lst = [1, 2, 3, 4, 5, 6]
even = []
for i in lst:
    if i % 2 == 0:
        even.append(i)


for i in range(len(even)):
    for j in range(i+1):
        print(even[j], end=" ")
    print()
```

Pattern 6: Character List Pattern

List: ['A','B','C']

A

A B

A B C

```
lst = ['A', 'B', 'C']
for i in range(1, len(lst)+1):
    for j in range(i):
        print(lst[j], end=" ")
    print()
```

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